Rachel Eglash, Peter Kudelin, Sena Lee
November 5, 2025
Math2B – Ryker

Project Proposal (Resubmit)

1. **The research question I will be answering using techniques from this class is:**
   Do different grade weighting methods produce more equitable outcomes across different demographic groups?
   Other related questions are:
   Which specific demographic groups does this apply to (gender, socioeconomic status, disability status, race, etc.)?
   How does including homework and "employability," (from Griffin's dataset, or a combination of participation, attendance, and behavior from the UCI dataset) change the final grades for students in different demographic groups?
   What is the magnitude of the equity gap (the difference in the mean grades) in different demographic groups under different grade weighting methods?
   Which demographic groups benefit the most from traditional grading that includes non-academic factors, and which are most disadvantaged?

2. **The techniques from this class that I will demonstrate to answer my research question are:**
   Weighted linear combinations and matrix-vector multiplication
   Student grades can be modeled as weighted linear combinations of multiple grade components. The different grade weighting methods represent different weights of vectors applied to the same group of students with their given grades (represented in a matrix). By calculating final grades using multiple different grade weighting methods using matrix-vector multiplication and comparing the equity gaps between different demographic groups, I can determine which grading method is most equitable (while still maintaining academic rigor requirements).
   The weekly learning objectives in context that it demonstrates are:
   Week 2 – Write a vector as a linear combination of a given set of vectors.
   Week 5 – Construct a basis for a vector space and determine if a given set forms a basis.
   The general form of the linear system that I'll be solving in my project is:
   $Ax = b$
   where given data from m total students, and given three scores (assessment, homework, and employability):
   - A is an $m \times 3$ matrix containing the three scores for each student,
   - x is a 3×1 vector containing the weights for the three scores for each student, and
   - b is an $m \times 1$ column vector containing the final grade for each student.

   In the traditional grade weighting method, which only takes tests into account, the weight for the assessment score is 1, the weight for the homework score is 0, and the weight for the employability score is 0. In different grade weighting methods, the weights take assessment score, homework score, and employability score into account, where the weights for the three score must add to 1.
   The connection to both the research question and theory from this class is:
   By computing final grades under multiple weighting methods and calculating the resulting equity gaps between different demographic groups, We can determine which method minimizes disparities. The method that produces the smallest equity gaps while maintaining academic standards is the most equitable method.

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3. **My group members are:**
   Rachel Eglash, Peter Kudelin, Sena Lee

4. **At least one source that I'll be using for your project (that clearly shows how to solve a similar type of linear system or otherwise connect to the mathematical model or claim used in this project to answer questions similar to my research question) is:**

   Main source:

   Griffin, Robert Thomas, "Grading and equity: Inflation/deflation based on race, gender, socio-economic and disability statuses when homework and employability scores are included" (2020). *Dissertations and Theses @ UNI*. 1045. https://scholarworks.uni.edu/etd/1045

   This source contains grades for math students with the grades separated into three categories (assessment, homework, employability) and includes demographic information such as race, gender, socioeconomic status, and disability status. Although this is a graduate-level paper, it is not a linear algebra graduate-level paper, therefore the linear algebra used to analyze the data is not at a level beyond the scope of this class.

   Additional possible source for data:

   Cortez, P. and A. M. Gonçalves Silva. "Using data mining to predict secondary school student performance." (2008). https://www.semanticscholar.org/paper/Using-data-mining-to-predict-secondary-school-Cortez-Silva/61d468d5254730bbecf822c6b60d7d6595d9889c

   This source contains grades for Portuguese math students and includes demographic information such as gender, parental education, health, and multiple grade components that can represent different assessment types.

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More Information about the Matrices:

$b = Ax$, where m is the number of students

$$A = \begin{bmatrix} a_1 & h_1 & e_1 \\ \vdots & \vdots & \vdots \\ a_i & h_i & e_i \\ \vdots & \vdots & \vdots \\ a_m & h_m & e_m \end{bmatrix}$$

where for student i, a_i is their assessment score,
h_i is their homework score, and
e_i is their employability score.

$$x = \begin{bmatrix} w_a \\ w_h \\ w_e \end{bmatrix}$$

where w_a is the weight for assessment scores,
w_h is the weight for homework scores, and
w_e is the weight for employability scores,
given that $w_a + w_h + w_e = 1$

$$b = \begin{bmatrix} g_1 \\ \vdots \\ g_i \\ \vdots \\ g_m \end{bmatrix}$$

where for student i, g_i is their final grade.

$$\begin{bmatrix} a_1 & h_1 & e_1 \\ \vdots & \vdots & \vdots \\ a_i & h_i & e_i \\ \vdots & \vdots & \vdots \\ a_m & h_m & e_m \end{bmatrix} \begin{bmatrix} w_a \\ w_h \\ w_e \end{bmatrix} = \begin{bmatrix} g_1 \\ \vdots \\ g_i \\ \vdots \\ g_m \end{bmatrix}$$